

WIPRO--JAVA SCRIPT-ASSIGNMENT-→

1.Lab Question: Alert, Confirm, and Prompt – Quiz Game

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Quiz Game</title>

<script>

    function startQuiz() {

        alert("Welcome to the Quiz Game!");

        let name = prompt("What's your name?");

        if (!name || name.trim() === "") {

            alert(" You didn't enter your name. Exiting game.");

            return;

        }

        let ready = confirm(`Hi ${name}! Are you ready for the quiz?`);

        if (!ready) {

            alert("Okay, maybe next time!");

            return;

        }

        let answer = prompt(

            "How many states are there in India?\n" +

            "a) 28\n" +
```

```
"b) 29\n" +  
"c) 26"  
);
```

```
if (!answer) {  
    alert("You didn't answer. Game over!");  
    return;  
}
```

```
answer = answer.trim().toLowerCase();  
if (answer === "a" || answer === "28") {  
    alert(`Correct! Well done, ${name} `);  
} else {  
    alert(" Oops! The correct answer is 28.");  
}  
}
```

```
</script>
```

```
</head>
```

```
<body onload="startQuiz()">
```

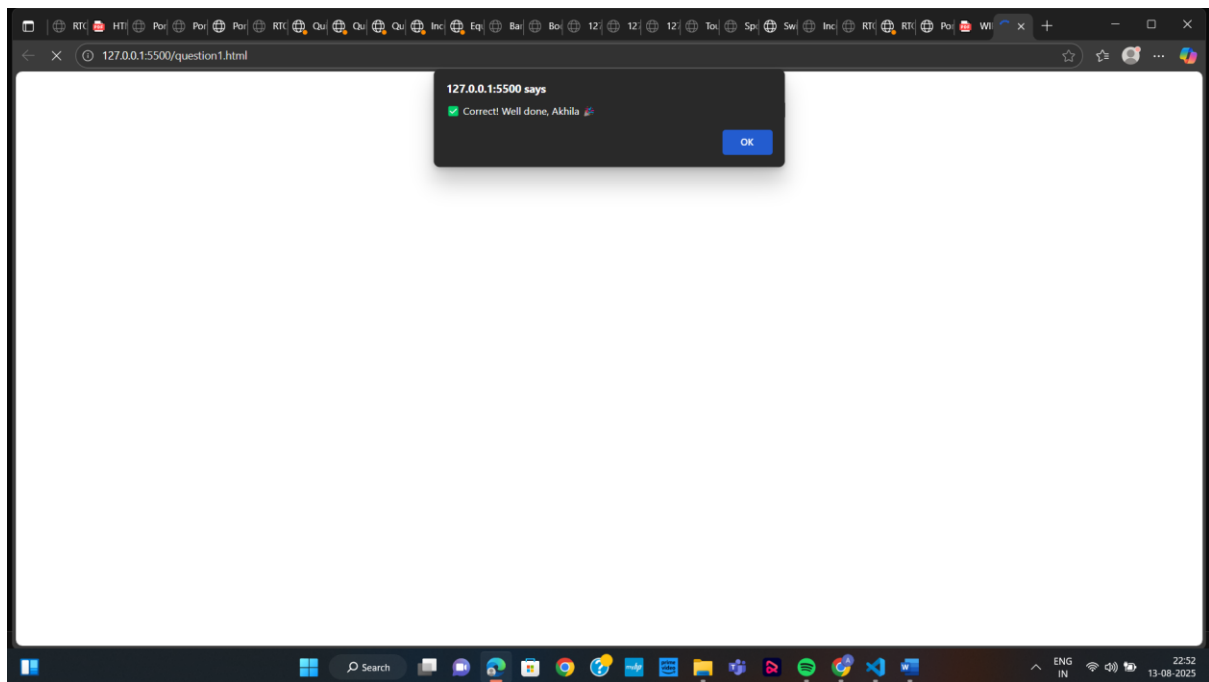
```
<h1>Simple Quiz Game</h1>
```

```
<p>Reload the page to play again.</p>
```

```
</body>
```

```
</html>
```

Output:



2.Lab Question: Income Tax Calculator

Source code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Income Tax Calculator</title>

<script>

function calculateTax() {

    let name = prompt("Enter your full name:");

    if (!name || name.trim() === "") {

        alert("Name cannot be empty!");
```

```

    return;
}

let pan = prompt("Enter your PAN number (10 characters, uppercase:");
if (!pan || pan.length !== 10 || pan !== pan.toUpperCase()) {
    alert("Invalid PAN format! It must be 10 characters and uppercase.");
    return;
}

let dobInput = prompt("Enter your Date of Birth (YYYY-MM-DD):");
let dob = new Date(dobInput);
if (isNaN(dob.getTime())) {
    alert("Invalid Date format!");
    return;
}

let today = new Date();
let age = today.getFullYear() - dob.getFullYear();
if (
    today.getMonth() < dob.getMonth() ||
    (today.getMonth() === dob.getMonth() && today.getDate() < dob.getDate())
) {
    age--;
}

let income = parseFloat(prompt("Enter your annual income (in ₹):"));
if (isNaN(income) || income < 0) {
    alert("Invalid income amount!");
    return;
}

let tax = 0;

```

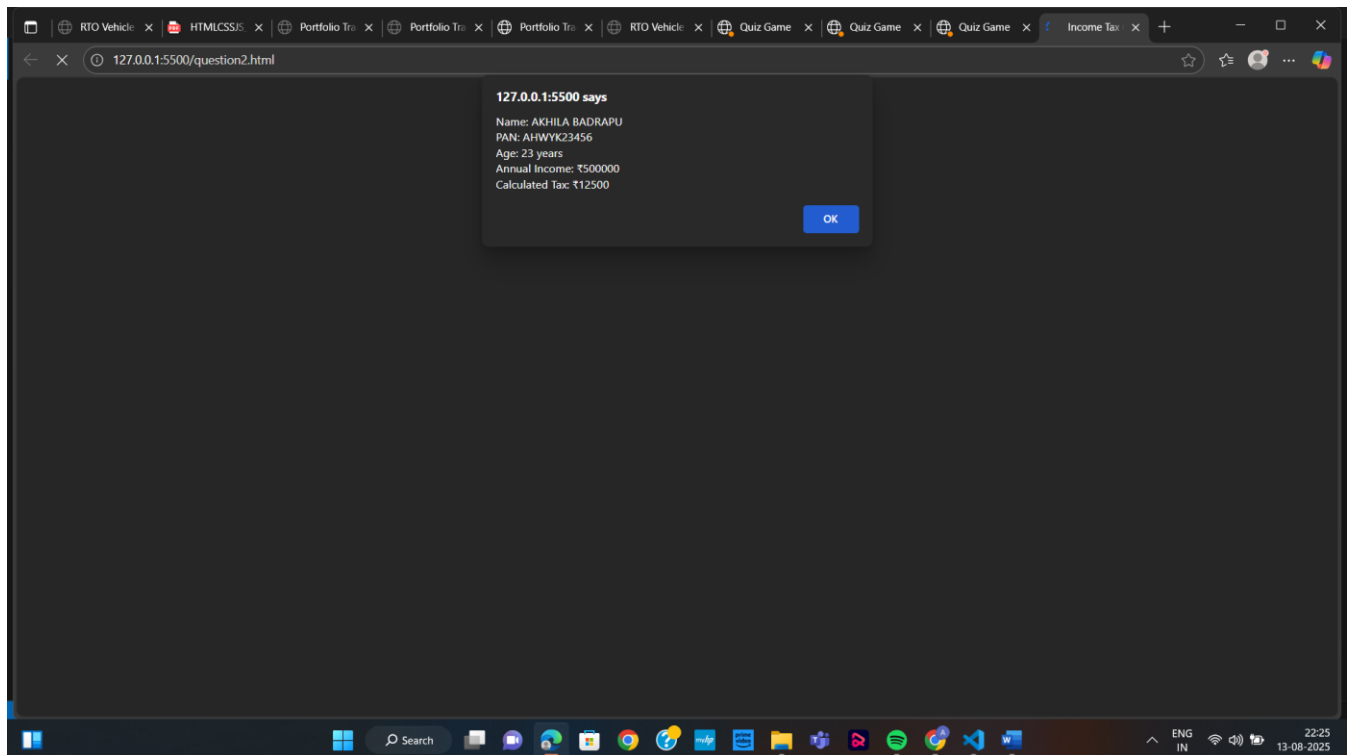
```
if (income <= 250000) {
    tax = 0;
} else if (income <= 500000) {
    tax = (income - 250000) * 0.05;
} else if (income <= 1000000) {
    tax = (250000 * 0.05) + ((income - 500000) * 0.2);
} else {
    tax = (250000 * 0.05) + (500000 * 0.2) + ((income - 1000000) * 0.3);
}

tax = Math.round(tax);

let result =
    "Name: " + name.toUpperCase() + "\n" +
    "PAN: " + pan + "\n" +
    "Age: " + age + " years\n" +
    "Annual Income: ₹" + income + "\n" +
    "Calculated Tax: ₹" + tax;

alert(result);
}
</script>
</head>
<body onload="calculateTax()">
<h1>Income Tax Calculator</h1>
<p>Reload the page to calculate again.</p>
</body>
</html>
```

Output:



3.Lab Question: Equality & Inequality Checker.

Source Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Equality & Inequality Checker</title>
```

```
<script>
```

```
function compareValues() {
```

```
    let inputOne = prompt("Enter the first value:");
```

```
let inputTwo = prompt("Enter the second value:");
```

```
let typeOne = typeof inputOne;
```

```
let typeTwo = typeof inputTwo;
```

```
let numberOne = Number(inputOne);
```

```
let numberTwo = Number(inputTwo);
```

```
let looseEqual = (inputOne == inputTwo);
```

```
let strictEqual = (inputOne === inputTwo);
```

```
let looseNotEqual = (inputOne != inputTwo);
```

```
let strictNotEqual = (inputOne !== inputTwo);
```

```
let output =
```

```
    "First Value: " + inputOne + " (" + typeOne + ")\n" +
```

```
    "Second Value: " + inputTwo + " (" + typeTwo + ")\n\n" +
```

```
    "Using == : " + looseEqual + " → Equality ignores type\n" +
```

```
    "Using === : " + strictEqual + " → Strict equality checks type too\n" +
```

```
    "Using != : " + looseNotEqual + " → Inequality ignores type\n" +
```

```
    "Using !== : " + strictNotEqual + " → Strict inequality checks type as well\n\n" +
```

```
    "Converted to Numbers for comparison:\n" +
```

```
    numberOne + " (" + typeof numberOne + "), " +
```

```
    numberTwo + " (" + typeof numberTwo + ")";
```

```
alert(output);
```

```
}
```

```
</script>
```

```
</head>
```

```

<body onload="compareValues()">

<h1>Equality & Inequality Checker</h1>

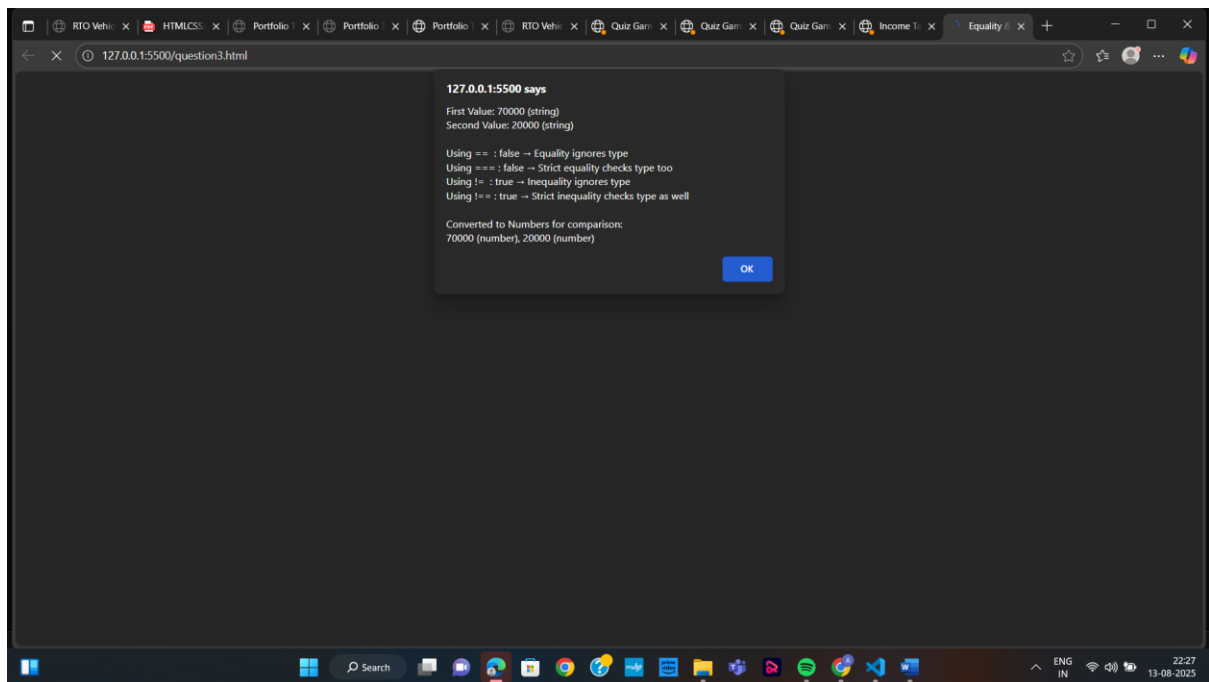
<p>Reload the page to compare again.</p>

</body>

</html>

```

Output:



4. Lab Question: Bank Customers – Dynamic Array Creation & Manipulation

Source code:

```

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Bank Customers Management</title>

<script>

```

```

function manageCustomers() let totalCustomers = parseInt(prompt("How many customers
to register?"));

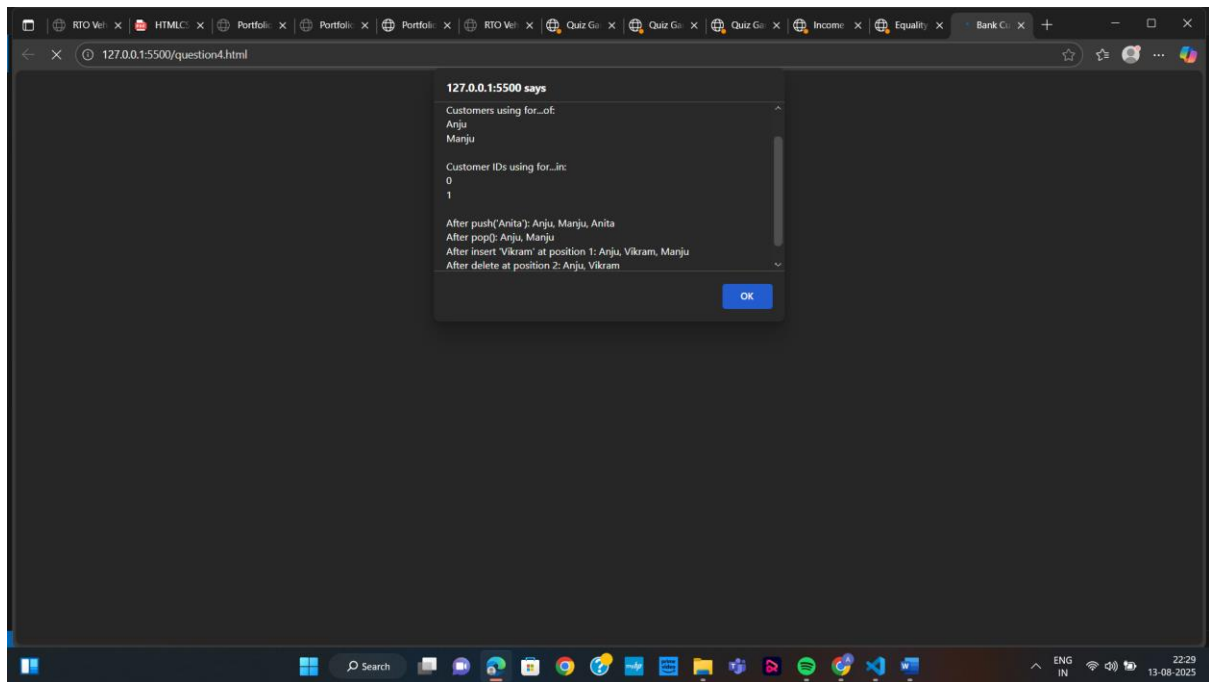
```



```
if (isNaN(totalCustomers) || totalCustomers <= 0) {  
    alert("Invalid number of customers!");  
    return;  
}  
  
let customers = [];  
  
for (let i = 0; i < totalCustomers; i++) {  
    let name = prompt("Enter name of customer " + (i + 1) + ":");  
    customers.push(name);  
}  
  
let namesList = "Customers using for...of:\n";  
  
for (let cust of customers) {  
    namesList += cust + "\n";  
}  
  
let idsList = "Customer IDs using for...in:\n";  
  
for (let index in customers) {  
    idsList += index + "\n";  
}  
  
customers.push("Anita");  
  
let afterPush = "After push('Anita'): " + customers.join(", ");  
  
customers.pop();  
  
let afterPop = "After pop(): " + customers.join(", ");  
  
customers.splice(1, 0, "Vikram");  
  
let afterInsert = "After insert 'Vikram' at position 1: " + customers.join(", ");  
  
customers.splice(2, 1);  
  
let afterDelete = "After delete at position 2: " + customers.join(", ");  
  
  
let finalList = "Final Customers: " + customers.join(", ");
```

```
    alert(
        "Original Customers: " + customers.join(", ") + "\n\n" +
        namesList + "\n" +
        idsList + "\n" +
        afterPush + "\n" +
        afterPop + "\n" +
        afterInsert + "\n" +
        afterDelete + "\n" +
        finalList
    );
}
</script>
</head>
<body onload="manageCustomers()">
<h1>Bank Customers – Dynamic Array Management</h1>
<p>Reload the page to run again.</p>
</body>
</html>
```

Output:



5. Lab Question: JSON – Book Store Management

Source code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Book Store Management</title>
```

```
<script>
```

```
function manageBooks() {
```

```
    let books = [
```

```
        { title: "HTML", author: "Akhila", price: 500, available: true },
```

```
        { title: "BootStrap", author: "Anjali", price: 450, available: false }
```

```
    ] let allBooks = "All Books:\n";
```

```

    for (let book of books) {

        allBooks += `Title: ${book.title}, Author: ${book.author}, Price: ₹${book.price}, Available:
${book.available}\n`;

    }

    let availableBooks = "Available Books:\n";

    for (let book of books) {

        if (book.available) {

            availableBooks += `Title: ${book.title}, Author: ${book.author}, Price:
₹${book.price}\n`;

        }

    }

    books.push({ title: "React.js", author: "Shankar", price: 600, available: true });

    let afterAdding = "After Adding:\n" + books.map(b => b.title).join(", ");

    for (let book of books) {

        if (book.title === "React") {

            book.price = 500;

        }

    }

    let afterPriceUpdate = "After Price Update:\n" + books.map(b => `${b.title} -
₹${b.price}`).join("\n");

    let indexToDelete = books.findIndex(b => b.title === "HTML");

    if (indexToDelete !== -1) {

        books.splice(indexToDelete, 1);

    }

    let afterDeletion = "After Deletion:\n" + books.map(b => b.title).join(", ");

    alert(allBooks + "\n" + availableBooks + "\n" + afterAdding + "\n\n" + afterPriceUpdate +
"\n\n" + afterDeletion);

}

</script>

```

```

</head>

<body onload="manageBooks()">

<h1>Book Store Management</h1>

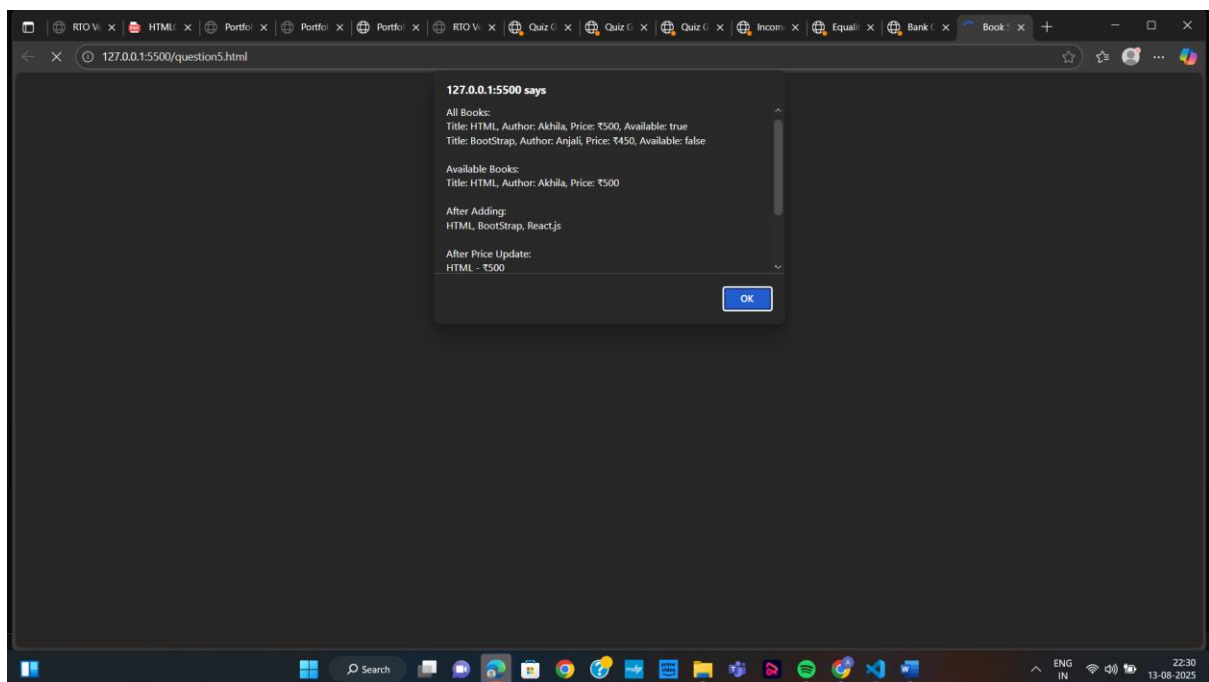
<p>Reload the page to run again.</p>

</body>

</html>

```

Output:



6. Lab Question: Tour Booking Form – getElementById()

Source Code:

```

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Tour Booking Form</title>

```

```
<style>
  body {
    font-family: Arial, sans-serif;
    margin: 20px;
  }
  label {
    display: block;
    margin-top: 10px;
  }
  input, select, button {
    padding: 8px;
    margin-top: 5px;
    width: 250px;
  }
  #summary {
    margin-top: 20px;
    padding: 15px;
    border: 1px solid #ccc;
    background: #f9f9f9;
    width: 300px;
  }
</style>

<script>
function bookTour() {
  let fullName = document.getElementById("name").value;
  let email = document.getElementById("email").value;
  let destination = document.getElementById("destination").value;
  let days = parseInt(document.getElementById("days").value);
```

```

    let travelers = parseInt(document.getElementById("travelers").value);
    if (!fullName || !email || isNaN(days) || isNaN(travelers) || days <= 0 || travelers <= 0) {
        alert("Please fill in all fields correctly.");
        return;
    }
    let pricePerDay = 0;
    if (destination === "Munnar") pricePerDay = 2000;
    else if (destination === "Arunachala temple") pricePerDay = 3000;
    else if (destination === "Wayand") pricePerDay = 4000;
    let totalCost = pricePerDay * days * travelers;
    let summary = `
        Name: ${fullName} <br>
        Email: ${email} <br>
        Destination: ${destination} <br>
        Days: ${days} <br>
        Travelers: ${travelers} <br>
        Total Price: ₹${totalCost}
    `;
    document.getElementById("summary").innerHTML = summary;
}

```

```

</script>

```

```

</head>

```

```

<body>

```

```

<h2>Tour Booking Form</h2>

```

```

<form onsubmit="event.preventDefault(); bookTour();">

```

```

    <label>Full Name:</label>

```

```

    <input type="text" id="name">

```

```
<label>Email:</label>
```

```
<input type="email" id="email">
```

```
<label>Destination:</label>
```

```
<select id="destination">
```

```
  <option value="Munnar">Munnar</option>
```

```
  <option value="Arunachala temple">Arunachala temple</option>
```

```
  <option value="Wayand">Wayan</option>
```

```
</select>
```

```
<label>Number of Days:</label>
```

```
<input type="number" id="days">
```

```
<label>Number of Travelers:</label>
```

```
<input type="number" id="travelers">
```

```
<br><br>
```

```
<button type="submit">Book Now</button>
```

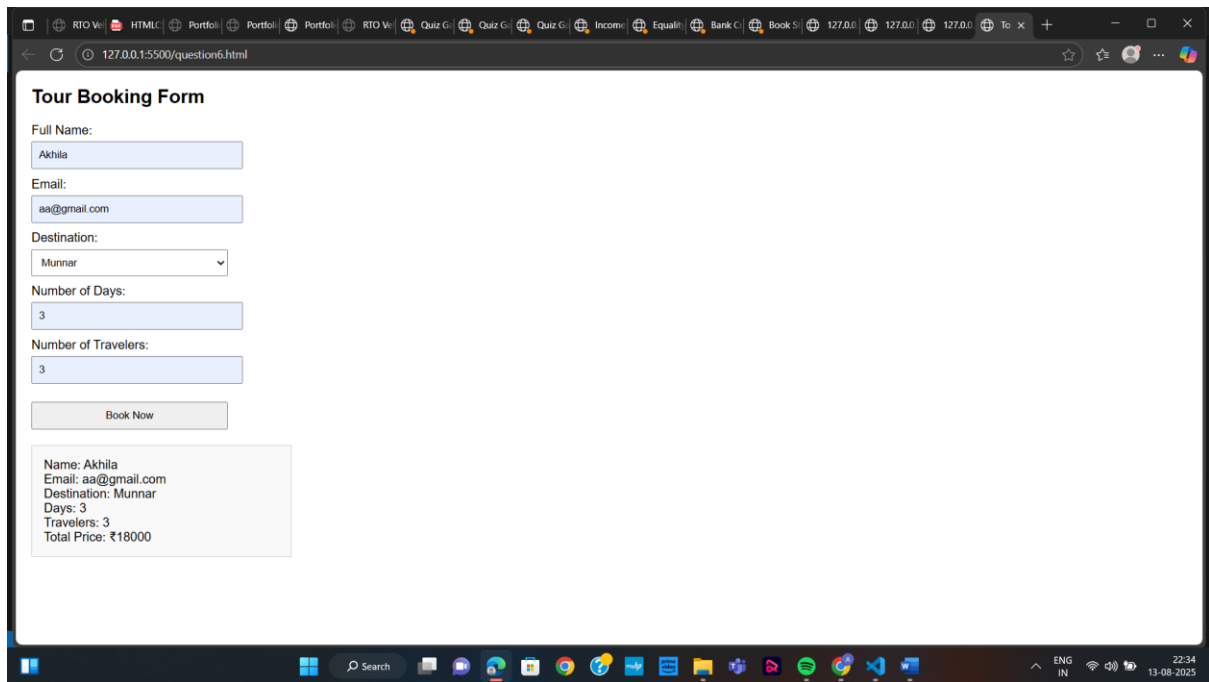
```
</form>
```

```
<div id="summary"></div>
```

```
</body>
```

```
</html>
```

Output:



7

7. Lab Question – Sports Team Player Details

Output:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Sports Team Player Details</title>
```

```
<script>
```

```
function showPlayers() {
```

```
    const player1 = {
```

```
        name: "Virat Kohli",
```

```
        age: 36,
```

```
        sport: "Cricket",
```

```
        stats: {
```

```
            matches: 500,
```

```

        runs: 25000,
        average: 57.2
    }
};

const player2 = {
    name: "Shubman Gill",
    age: 38,
    sport: "Cricket",
    stats: {
        matches: 200,
        runs: 10000,
        average: 45
    }
};

const players = [player1, player2];

const [p1, p2] = players;

const { name: name1, sport: sport1, stats: { matches: matches1, runs, average } } = p1;
//Destructure Player 2 (Cricket)
const { name: name2, sport: sport2, stats: { matches: matches2, goals, assists } } = p2;

// Output

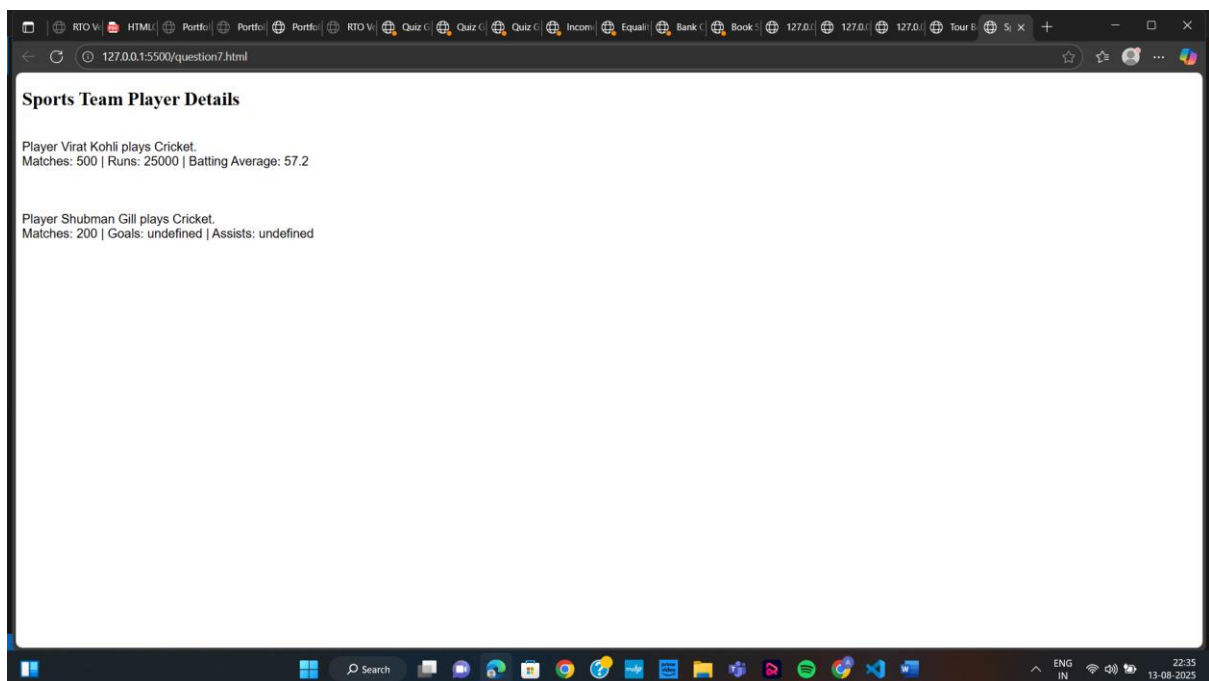
let output = `
    Player ${name1} plays ${sport1}.
    Matches: ${matches1} | Runs: ${runs} | Batting Average: ${average}
    <br><br>
    Player ${name2} plays ${sport2}.
    Matches: ${matches2} | Goals: ${goals} | Assists: ${assists}
`;

```

```
// Display on page

document.getElementById("result").innerHTML = output;
}
</script>
</head>
<body onload="showPlayers()">
<h2>Sports Team Player Details</h2>
<div id="result" style="white-space: pre-line; font-family: Arial;"></div>
</body>
</html>
```

Output:



8.Lab Question – Switchboard Control (let, var, const)

Source Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Switchboard Control</title>

<script>

function controlSwitchboard() {

    const buildingName = "Sunrise Apartments";

    let status = "OFF";

    const devices = ["Light", "Fan", "AC"];

    for (var i = 0; i < devices.length; i++) {

        status = (status === "OFF") ? "ON" : "OFF";

        console.log(`Checking Switch ${i + 1}: ${devices[i]} is ${status}`);

    }

    console.log(`All switches checked for ${buildingName}.`);

    console.log(`Outside loop: var counter is ${i}`);

    try {

        buildingName = "Moonlight Residency";

    } catch (err) {

        console.error("Error when trying to change building name:", err.message);

    }

}

</script>

</head>

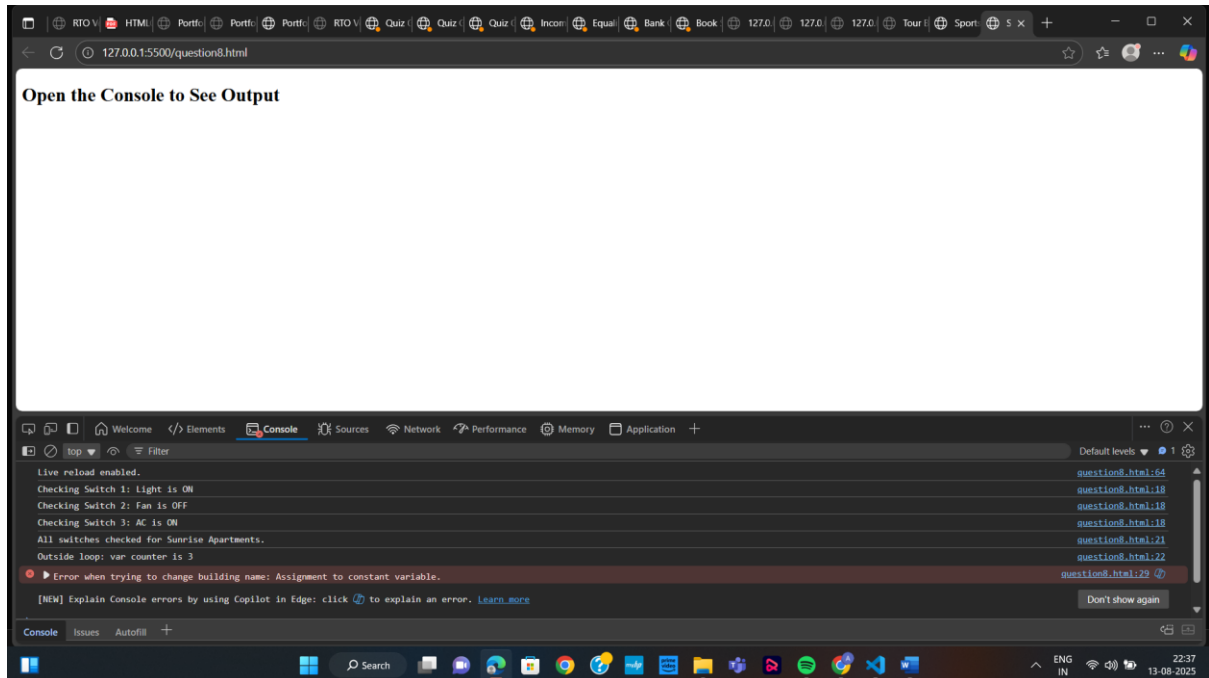
<body onload="controlSwitchboard()">

<h2>Open the Console to See Output</h2>

</body>
```

</html>

Output:



9. .Lab Question: Income Tax Calculator with Multiple Callbacks

Source Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Income Tax Calculator (Multiple Callbacks)</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Income Tax Calculator (Multiple Callbacks)</h1>
```

```
  <div id="result"></div>
```

```
  <script> let taxRecords = [];
```

```

function calculateTax(name, pan, income, callback) {
    let tax = 0;
    if (income <= 250000) {
        tax = 0;
    } else if (income <= 500000) {
        tax = income * 0.05;
    } else if (income <= 1000000) {
        tax = income * 0.20;
    } else {
        tax = income * 0.30;
    }
    tax = Math.round(tax);
    callback(name, pan, income, tax);
}

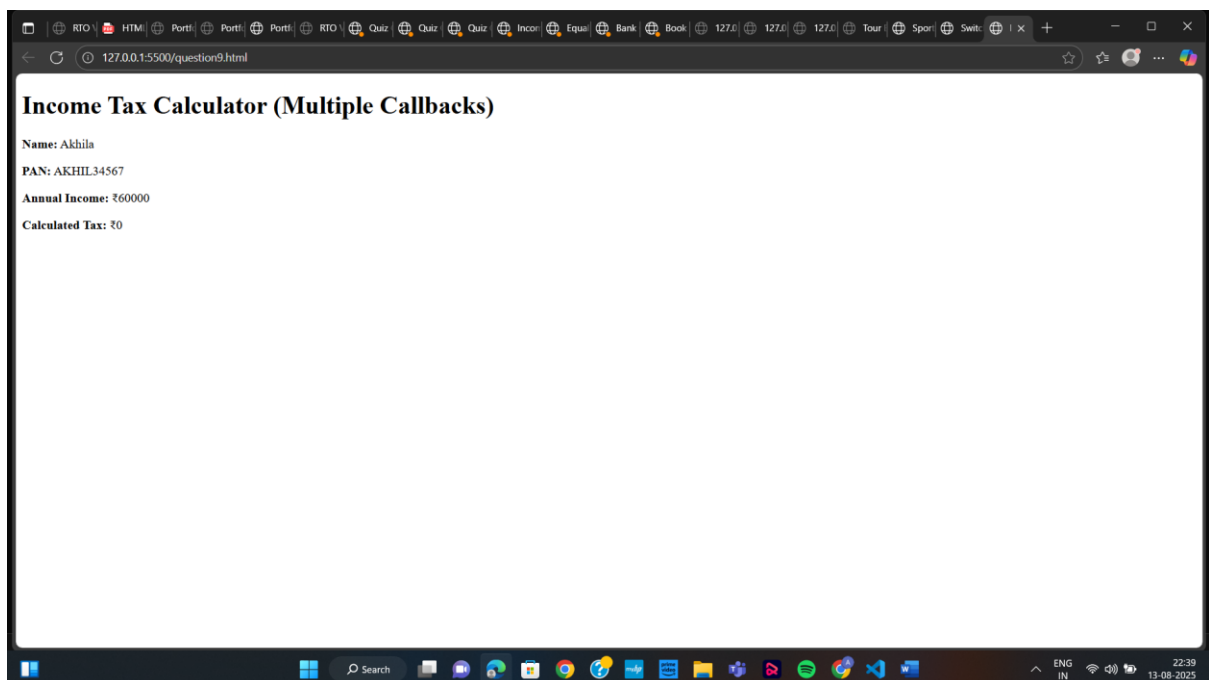
function logToConsole(name, pan, income, tax) {
    console.log(`Name: ${name}`);
    console.log(`PAN: ${pan}`);
    console.log(`Annual Income: ₹${income}`);
    console.log(`Calculated Tax: ₹${tax}`);
}

function displayOnPage(name, pan, income, tax) {
    document.getElementById("result").innerHTML = `
        <p><strong>Name:</strong> ${name}</p>
        <p><strong>PAN:</strong> ${pan}</p>
        <p><strong>Annual Income:</strong> ₹${income}</p>
        <p><strong>Calculated Tax:</strong> ₹${tax}</p>
    `
}

```

```
function saveToRecords(name, pan, income, tax) {  
    taxRecords.push({ name, pan, income, tax });  
    console.log("Saved Records:", taxRecords);  
}  
  
let name = prompt("Enter Name:");  
let pan = prompt("Enter PAN:");  
let income = Number(prompt("Enter Annual Income:"));  
calculateTax(name, pan, income, logToConsole);  
calculateTax(name, pan, income, displayOnPage);  
calculateTax(name, pan, income, saveToRecords);  
  
</script>  
</body>  
</html>
```

Output:



10. Build a multi-section RTO Vehicle Registration form with Bootstrap that validates:

Source Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>RTO Vehicle Registration</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">

<style>

    .progress {

        height: 20px;

    }

</style>

</head>

<body class="bg-light">

<div class="container mt-4 mb-5">

    <h2 class="mb-4 text-center">RTO Vehicle Registration Form</h2>

    <!-- Progress Bar -->

    <div class="progress mb-4">

        <div id="formProgress" class="progress-bar bg-success" role="progressbar"
style="width: 0%">0%</div>

    </div>

    <form id="rtoForm" class="needs-validation" novalidate>

        <!-- Accordion -->
```



```
<div class="accordion" id="formAccordion">

<!-- Section 1: Owner -->

<div class="accordion-item">

    <h2 class="accordion-header">

        <button class="accordion-button" type="button" data-bs-toggle="collapse" data-
bs-target="#ownerSection" aria-expanded="true">

            1. Owner Details

        </button>

    </h2>

    <div id="ownerSection" class="accordion-collapse collapse show" data-bs-
parent="#formAccordion">

        <div class="accordion-body row g-3">

            <div class="col-md-6">

                <label class="form-label">Full Name</label>

                <input type="text" class="form-control" name="fullName" placeholder="e.g.
Priya Natraajan" required>

                <div class="invalid-feedback">Please enter full name.</div>

            </div>

            <div class="col-md-6">

                <label class="form-label">Date of Birth</label>

                <input type="date" class="form-control" name="dob" required>

                <div class="invalid-feedback">You must be at least 18 years old.</div>

            </div>

            <div class="col-md-6">

                <label class="form-label">Email</label>

                <input type="email" class="form-control" name="email"
placeholder="name@example.com" required>

                <div class="invalid-feedback">Enter a valid email.</div>

            </div>

        </div>

    </div>

</div>
```

```
<div class="col-md-6">

    <label class="form-label">Mobile (India)</label>

    <input type="tel" class="form-control" name="mobile"
pattern="\^+91\d{10}$" placeholder="+91 98XXXXXXXX" required>

    <div class="invalid-feedback">Enter a valid Indian mobile number
(+91XXXXXXXXXX).</div>

</div>

<div class="col-md-6">

    <label class="form-label">Aadhaar</label>

    <input type="text" class="form-control" name="aadhaar"
pattern="\^d{4}\sd{4}\sd{4}$" placeholder="1334 5679 9012" required>

    <div class="invalid-feedback">Enter a valid 12-digit Aadhaar number.</div>

</div>

<div class="col-md-6">

    <label class="form-label">PAN</label>

    <input type="text" class="form-control" name="pan" pattern="\^[A-Z]{5}[0-
9]{4}[A-Z]{1}$" placeholder="ABCDE1234F" required>

    <div class="invalid-feedback">Enter a valid PAN number.</div>

</div>

<div class="col-md-6">

    <label class="form-label">Driving License No.</label>

    <input type="text" class="form-control" name="dl" placeholder="Format:
State(2) + RTO(2) + 11 digits" required>

    <div class="invalid-feedback">Enter a valid Driving License number.</div>

</div>

</div>

</div>

</div>
```

```
<!-- Section 2: Address -->

<div class="accordion-item">

  <h2 class="accordion-header">

    <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#addressSection">

      2. Address

    </button>

  </h2>

  <div id="addressSection" class="accordion-collapse collapse" data-bs-
parent="#formAccordion">

    <div class="accordion-body row g-3">

      <div class="col-md-12">

        <label class="form-label">Address Line 1</label>

        <input type="text" class="form-control" name="address1" required>

        <div class="invalid-feedback">Enter address.</div>

      </div>

      <div class="col-md-6">

        <label class="form-label">City</label>

        <input type="text" class="form-control" name="city" required>

        <div class="invalid-feedback">Enter city.</div>

      </div>

      <div class="col-md-6">

        <label class="form-label">State</label>

        <select class="form-select" name="state" required>

          <option value="">Choose...</option>

          <option>Tamil Nadu</option>

          <option>Karnataka</option>

          <option>Maharashtra</option>

        </select>

      </div>

    </div>

  </div>

</div>
```

```
        <div class="invalid-feedback">Select a state.</div>
    </div>
    <div class="col-md-6">
        <label class="form-label">PIN Code</label>
        <input type="text" class="form-control" name="pin" pattern="^\d{6}$"
required>
        <div class="invalid-feedback">Enter valid 6-digit PIN code.</div>
    </div>
</div>
</div>
</div>
</div>
```

```
<!-- Section 3: Vehicle -->
<div class="accordion-item">
    <h2 class="accordion-header">
        <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#vehicleSection">
            3. Vehicle Details
        </button>
    </h2>
    <div id="vehicleSection" class="accordion-collapse collapse" data-bs-
parent="#formAccordion">
        <div class="accordion-body row g-3">
            <div class="col-md-6">
                <label class="form-label">Registration No.</label>
                <input type="text" class="form-control" name="regNo" required>
                <div class="invalid-feedback">Enter registration number.</div>
            </div>
            <div class="col-md-6">
```

```
<label class="form-label">Vehicle Class</label>
<select class="form-select" name="vehicleClass" required>
  <option value="">Choose...</option>
  <option>Car</option>
  <option>Bike</option>
  <option>Truck</option>
</select>
<div class="invalid-feedback">Select vehicle class.</div>
</div>
<div class="col-md-6">
  <label class="form-label">Fuel Type</label>
  <select class="form-select" name="fuel" required>
    <option value="">Choose...</option>
    <option>Petrol</option>
    <option>Diesel</option>
    <option>Electric</option>
  </select>
  <div class="invalid-feedback">Select fuel type.</div>
</div>
<div class="col-md-6">
  <label class="form-label">Make</label>
  <input type="text" class="form-control" name="make" required>
  <div class="invalid-feedback">Enter make.</div>
</div>
<div class="col-md-6">
  <label class="form-label">Engine No.</label>
  <input type="text" class="form-control" name="engineNo" pattern="^[a-zA-Z0-9]{6,12}$" required>
```

```
        <div class="invalid-feedback">Enter valid engine number.</div>
    </div>
    <div class="col-md-6">
        <label class="form-label">Year of Manufacture</label>
        <input type="number" class="form-control" name="year" min="1900"
max="2025" required>
        <div class="invalid-feedback">Enter valid year.</div>
    </div>
    <div class="col-md-6">
        <label class="form-label">Purchase Date</label>
        <input type="date" class="form-control" name="purchaseDate" required>
        <div class="invalid-feedback">Date cannot be in the future.</div>
    </div>
    <div class="col-md-6">
        <label class="form-label">Color</label>
        <input type="text" class="form-control" name="color" required>
        <div class="invalid-feedback">Enter color.</div>
    </div>
</div>
</div>
</div>
```

```
<!-- Section 4: Compliance -->
<div class="accordion-item">
    <h2 class="accordion-header">
        <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#complianceSection">
            4. Compliance & Insurance
        </button>
```

</h2>

<div id="complianceSection" class="accordion-collapse collapse" data-bs-parent="#formAccordion">

<div class="accordion-body row g-3">

<div class="col-md-6">

<label class="form-label">Emission Norm</label>

<select class="form-select" name="emission" required>

<option value="">Choose...</option>

<option>BS-III</option>

<option>BS-IV</option>

<option>BS-VI</option>

</select>

<div class="invalid-feedback">Select emission norm.</div>

</div>

<div class="col-md-6">

<label class="form-label">Insurance Policy No.</label>

<input type="text" class="form-control" name="policyNo" required>

<div class="invalid-feedback">Enter policy number.</div>

</div>

<div class="col-md-6">

<label class="form-label">Insurer</label>

<input type="text" class="form-control" name="insurer" required>

<div class="invalid-feedback">Enter insurer name.</div>

</div>

<div class="col-md-6">

<label class="form-label">Insurance Expiry</label>

<input type="date" class="form-control" name="insuranceExpiry" required>

<div class="invalid-feedback">Date must be today or later.</div>

```

        </div>

    </div>

</div>

</div>

<!-- Section 5: Finance -->

<div class="accordion-item">

    <h2 class="accordion-header">

        <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#financeSection">

            5. Finance (Hypothecation)

        </button>

    </h2>

    <div id="financeSection" class="accordion-collapse collapse" data-bs-
parent="#formAccordion">

        <div class="accordion-body row g-3">

            <div class="col-md-12">

                <div class="form-check">

                    <input type="checkbox" class="form-check-input" id="hypothecation"
name="hypothecation">

                        <label class="form-check-label">Vehicle is under hypothecation /
loan</label>

                    </div>

                </div>

                <div id="financeDetails" class="row g-3 mt-2 d-none">

                    <div class="col-md-6">

                        <label class="form-label">Bank/NBFC</label>

                        <input type="text" class="form-control" name="bank">

                    </div>

```



```

        <div class="col-md-6">

            <label class="form-label">Agreement Date</label>

            <input type="date" class="form-control" name="agreementDate">

        </div>

    </div>

</div>

</div>

</div>

```

```

<!-- Section 6: Uploads -->

```

```

<div class="accordion-item">

    <h2 class="accordion-header">

        <button class="accordion-button collapsed" type="button" data-bs-
toggle="collapse" data-bs-target="#uploadSection">

            6. Uploads

        </button>

    </h2>

    <div id="uploadSection" class="accordion-collapse collapse" data-bs-
parent="#formAccordion">

        <div class="accordion-body row g-3">

            <div class="col-md-6">

                <label class="form-label">ID/Address Proof</label>

                <input type="file" class="form-control" name="idProof"
accept=".pdf,.jpg,.jpeg,.png" required>

                <div class="invalid-feedback">Upload valid file (PDF/JPG/PNG ≤ 2 MB).</div>

            </div>

            <div class="col-md-6">

                <label class="form-label">Insurance Copy</label>

```

```

        <input type="file" class="form-control" name="insuranceCopy"
accept=".pdf,.jpg,.jpeg,.png" required>

        <div class="invalid-feedback">Upload valid file (PDF/JPG/PNG ≤ 2 MB).</div>

    </div>

    <div class="col-md-6">

        <label class="form-label">Vehicle Image (optional)</label>

        <input type="file" class="form-control" name="vehicleImage"
accept=".jpg,.jpeg,.png">

    </div>

</div>

</div>

<!-- Terms & CAPTCHA -->

<div class="form-check mt-3">

    <input type="checkbox" class="form-check-input" name="terms" required>

    <label class="form-check-label">I agree to the terms and conditions</label>

    <div class="invalid-feedback">You must agree before submitting.</div>

</div>

<div class="mt-3">

    <label>CAPTCHA: <span id="captchaQuestion"></span></label>

    <input type="number" class="form-control" name="captcha" required>

    <div class="invalid-feedback">Please solve the CAPTCHA.</div>

</div>

```

```
        <button type="submit" id="submitBtn" class="btn btn-primary mt-4"
disabled>Submit</button>
```

```
    </form>
```

```
</div>
```

```
<!-- Toast -->
```

```
<div class="position-fixed bottom-0 end-0 p-3" style="z-index: 11">
```

```
    <div id="successToast" class="toast align-items-center text-bg-success border-0"
role="alert">
```

```
        <div class="d-flex">
```

```
            <div class="toast-body">
```

```
                Application submitted successfully!
```

```
            </div>
```

```
            <button type="button" class="btn-close btn-close-white me-2 m-auto" data-bs-
dismiss="toast"></button>
```

```
        </div>
```

```
    </div>
```

```
</div>
```

```
<script
```

```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></scrip
t>
```

```
<script>
```

```
document.addEventListener("DOMContentLoaded", function() {
```

```
    const form = document.getElementById("rtoForm");
```

```
    const progressBar = document.getElementById("formProgress");
```

```
    const submitBtn = document.getElementById("submitBtn");
```

```
    const captchaQ = document.getElementById("captchaQuestion");
```

```
let captchaAnswer = 0;
```

```
function generateCaptcha() {
```

```
    const a = Math.floor(Math.random() * 10) + 1;
```

```
    const b = Math.floor(Math.random() * 10) + 1;
```

```
    captchaAnswer = a + b;
```

```
    captchaQ.textContent = `${a} + ${b} = ?`;
```

```
}
```

```
generateCaptcha();
```

```
// Show finance details when checkbox checked
```

```
document.getElementById("hypothecation").addEventListener("change", function() {
```

```
    document.getElementById("financeDetails").classList.toggle("d-none", !this.checked);
```

```
});
```

```
form.addEventListener("input", function() {
```

```
    let validSections = 0;
```

```
    document.querySelectorAll(".accordion-item").forEach(section => {
```

```
        if (section.querySelectorAll(":invalid").length === 0) {
```

```
            validSections++;
```

```
        }
```

```
    });
```

```
    const percent = Math.floor((validSections / document.querySelectorAll(".accordion-item").length) * 100);
```

```
    progressBar.style.width = percent + "%";
```

```
    progressBar.textContent = percent + "%";
```

```
    submitBtn.disabled = !form.checkValidity();
```

```
});
```

```
form.addEventListener("submit", function(e) {
    e.preventDefault();
    if (parseInt(form.captcha.value) !== captchaAnswer) {
        form.captcha.setCustomValidity("Invalid captcha");
        form.captcha.reportValidity();
        return;
    } else {
        form.captcha.setCustomValidity("");
    }

    if (!form.checkValidity()) {
        e.stopPropagation();
    } else {
        const toast = new bootstrap.Toast(document.getElementById("successToast"));
        toast.show();
        form.reset();
        generateCaptcha();
        progressBar.style.width = "0%";
        progressBar.textContent = "0%";
        submitBtn.disabled = true;
    }
    form.classList.add("was-validated");
});
});
</script>
</body>
</html>
```

Output:

RTO Vehicle Registration Form

1. Owner Details

Full Name

e.g. Priya Natraajan

Date of Birth

dd-mm-yyyy

Email

name@example.com

Mobile (India)

+91 98XXXXXXX

Aadhaar

1334 5679 9012

PAN

ABCDE1234F

Driving License No.

Format: State(2) + RTO(2) + 11 digits

2. Address

3. Vehicle Details

4. Compliance & Insurance

5. Finance (Hypothecation)

6. Uploads

☐ I agree to the terms and conditions

CAPTCHA: 3 + 4 = ?

Submit

11. Lab Question – JavaScript Closure (Share Market Example)

Source Code:

Question11.js -> js file

```
function createPortfolioTracker() {  
  
    let portfolio = [];  
  
    function buyShare(company, quantity, pricePerShare) {  
        portfolio.push({ company, quantity, pricePerShare });  
        console.log(`Bought ${quantity} shares of ${company} at ₹${pricePerShare} each.`);  
    }  
  
    function sellShare(company, quantity) {  
        let found = false;
```

```

    for (let share of portfolio) {
        if (share.company === company) {
            if (share.quantity >= quantity) {
                share.quantity -= quantity;
                console.log(`Sold ${quantity} shares of ${company}.`);
            } else {
                console.log(`Not enough shares to sell for ${company}.`);
            }
            found = true;
            break;
        }
    }
    if (!found) {
        console.log(`No shares found for ${company}.`);
    }
}

function totalValue() {
    let total = 0;
    for (let share of portfolio) {
        total += share.quantity * share.pricePerShare;
    }
    console.log(`Portfolio Value: ₹${total}`);
}

// Public methods
return {
    buyShare,
    sellShare,

```

```
        totalValue
    };
}
const myPortfolio = createPortfolioTracker();
myPortfolio.buyShare("TCS", 10, 3500);
myPortfolio.buyShare("Infosys", 5, 1500);
myPortfolio.sellShare("TCS", 3);
myPortfolio.totalValue();
```

HTML FILE → index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Portfolio Tracker</title>
</head>
<body>
    <script src="question11.js"></script>
</body>
</html>
```

Output:

